

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
<https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1>
3. Khan Academy, Compound Circuits:
<https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power>
4. EEVBlog, Voltage Dividers:
https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWIH0UOZNtOn-FIFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
<https://www.youtube.com/watch?v=WBfAEeEzDlq&list=PLvOlSehNtuHtWIH0UOZNtOn-FIFCn1GYw&index=13>

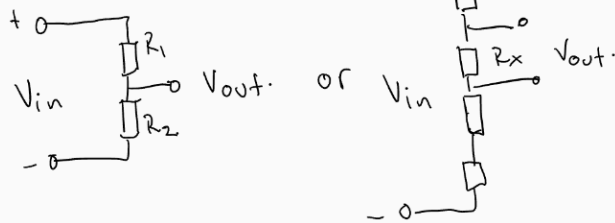
Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
 - a. [Lecture 5: Intro to DC/DC, Part 1](#)
2. [Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic](#)
3. [Power Electronics by Daniel W. Hart](#)

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Voltage dividers

→ Resistor V.D

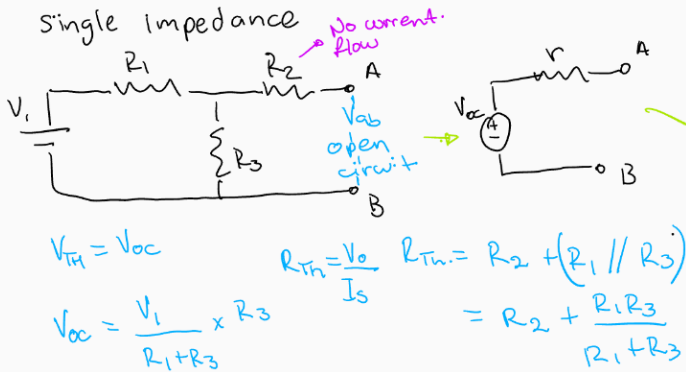


$$V_{out} = \frac{R_x}{R_{total}} \cdot V_{in} ; R_x = \text{ref. resistor.}$$

Voltage Source

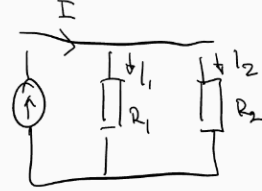
→ Thevenin's Theorem.

Any linear network that has V_s, I_s, R only can be reduced to a single source and single impedance



- To look for V_{th} , remove the R_L (creating V_{oc}).
 V at the node A = V at the node between R_1 and R_3 .
- To look for R_{th} , remove V_s by shorting the points across it.

→ Current V.D



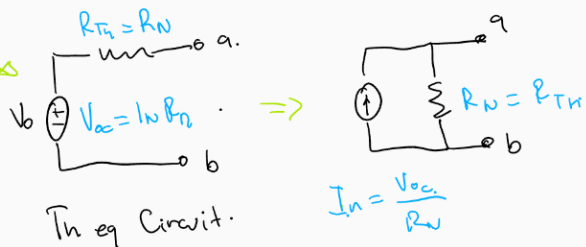
Use opp R

$$I_1 = \frac{R_2}{R_1 + R_2} \cdot I$$

only works for 2 resistors.

→ Norton's Theorem.

Any linear circuit containing V_s, I_s, R can be replaced by an equivalent circuit that consist of two components → an ideal current source in parallel with a resistor.



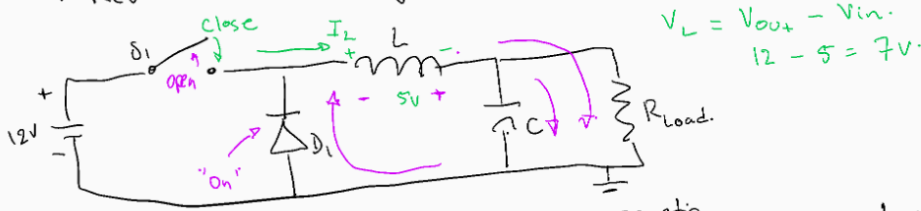
Th eq Circuit.

- Find R_N , replace all V_s with short circuits and all I_s with open ckts (nothing in that place)
 $\hookrightarrow$ combine all $R_{1,2,3,\dots}$ into 1.
- Find I_N , short the terminals a and b.
 $\hookrightarrow$ use nodal analysis to find I_N .

Example to understand the general idea of a buck (step-down) converter.

Buck converter

→ Reduces the voltage at the input, while it increases the current.



→ When $S_1 \rightarrow$ close, I_L increases $\rightarrow B$ (magnetic field) increases $\rightarrow L$ stores E
 $\rightarrow$ charges the capacitor
 $\rightarrow R_L$ receives I_{Bat}

→ When $S_1 \rightarrow$ open, I_L decreases $\rightarrow B \downarrow \rightarrow$ this causes to induce V opposite polarity and becomes a current source.
 $\rightarrow R_L$ receives I_L

- If $V_L > V_C$, it can charge the capacitor.
- if $V_L = V_C$, both L & C can discharge thru R_{Load} .

• $V_{out} \rightarrow V_i$
 • Duty Cycle
 $D = \frac{t_{on}}{T} \rightarrow D = \frac{1ms}{4ms} = .25$
 T Period = 25%

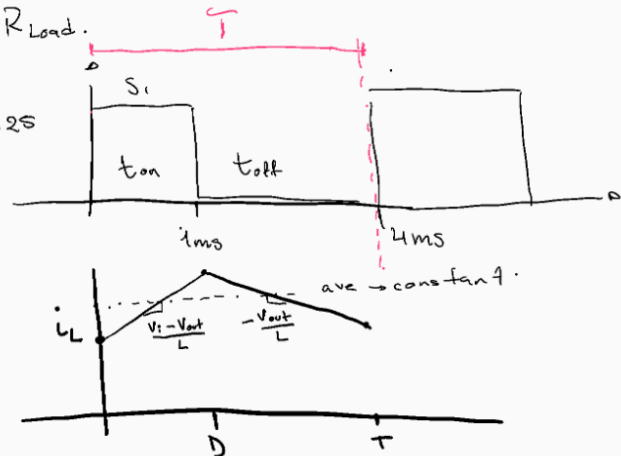
$\rightarrow V_{out} = V_i(D) = 12V(.25) = 3V$

Assuming: CCM and SS (average)

$\frac{V_i - V_{out}}{L} DT = -(-\frac{V_{out}}{L})(1-D)T$ most be true

$DV_i - DV_{out} = V_{out} - DV_{out}$

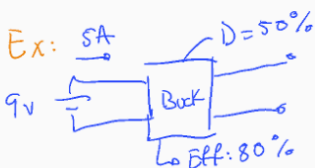
$V_{out} = V_i(D)$



• CCM = Continuous Conduction mode: L continuously feeds i to the load. i_L never falls to zero during the entire switching cycle.

• SS (ave) - Steady State: value of the dynamic elements are constant.

* We can adjust V_{out} by adjusting the duty cycle.



$P_{in} = VI = 9 \cdot 5 = 45W$
 $V_{out} = V_i(D) = 9(.50) = 4.5V$

$P_{out} = 36W$
 $P_{out} = VI = I_{out} = \frac{P}{V} = \frac{36W}{4.5V} = 8A$

$5A \rightarrow 8A \Rightarrow i$ increase by a factor of 2.
 $9V \rightarrow 4.5 \Rightarrow v$ decrease by a factor of 1.6

Calculate values for converter

Duty cycle: $V_{out} = V_{in}(D) \Rightarrow D = \frac{V_{out}}{V_{in}} = \frac{5V}{13.5V} = 0.37 \rightarrow 37\%$

Inductance value: $L = \left(\frac{V_{in} - V_{out}}{\Delta i_L f_{sw}} \right) D = \frac{V_{out}(1-D)}{\Delta i_L f_{sw}}$, $\Delta i_L = L$ ripple current
(A peak to peak)

Ripple i

$$\Delta i_L = r \times I_{out_max}$$

$r = 0.2$ to 0.4 ripple ratio

$$\Delta i_L = 0.3 \times 2A = 0.6A$$

$$L = \frac{(13.5 - 5) \times 0.37}{0.6 \times (5 \times 10^5)} = 1.05 \times 10^{-5} H$$
$$= 10.5 \mu H$$

Saturation i

$$i_{peak} = i_{out} + \Delta i_L / 2$$

$$= 2A + 0.6/2 = 2.3A$$

Capacitor

$$C_{out} = \frac{\Delta i_L}{8 \times f_{sw} \times \Delta V_C} = \frac{0.6}{8 \times (5 \times 10^5) \times (0.01)} = 1.5 \times 10^{-5}$$
$$15 \mu F$$